

Ridha Ouni

Associate Data Scientist at Infor | Machine Learning & AI Specialist

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Data Scientist with a strong engineering foundation and hands-on experience in statistical modeling, deep learning, and federated learning. Currently working at Infor, where I design and optimize forecasting models to improve decision-making and operational efficiency. Trained as a Telecommunications Engineer and holder of dual Master's degrees in Data Science and Information Processing, I bring both technical depth and practical problem-solving skills. Recognized for adaptability, teamwork, and analytical thinking to deliver impactful, data-driven solutions.

Professional Experience

Associate Data Scientist, Infor Tunisia — 04-2025 – present, Tunis, Tunisia
- Developed statistical and machine learning models for demand forecasting.
- Implemented optimization algorithms to enhance forecast accuracy.
- Collaborated closely with the team to drive improvements and ensure alignment.

AI Research Intern (Master's Degree Thesis), Tunisia Polytechnic School — 08-2024 – 10-2024, Tunis, Tunisia
- Utilized deep learning models and federated learning techniques for early diagnosis of diabetic retinopathy.
- Leveraged transfer learning in CNN-based architectures.
- Applied various aggregation strategies, including FedAvg and FedProx.
- Integrated AI explainability tools like LIME and SHAP to improve model transparency and interpretability.

Data Science intern (End of Studies), Infor Tunisia — 02-2024 – 06-2024, Tunis, Tunisia
- Developed a controller to automate forecasting engine selection using Java.
- Collaborated with team members and provided recommendations for product improvements.
- Conducted thorough testing and validation of the forecasting models to ensure accuracy and reliability in predictions.

Software Engineering Intern, Talan — 06-2023 – 08-2023, Tunis, Tunisia
- Spearheaded the 'Bricolini' project, focusing on reducing DIY waste and promoting environmental sustainability
- Spearheaded the design and development of a cross-platform mobile application using the Flutter framework.
- Implemented advanced computer vision techniques, such as object detection and image classification, with the best model achieving an accuracy of 83%.

Summer Intern, Tunisie Télécom — 08-2022 – 08-2022, Medenine, Tunisia
- Recovered and resolved nearly 95% of ADSL disturbances, ensuring minimal downtime and improving service reliability.
- Gained hands-on experience with optical fiber technology by performing key operational tasks related to fiber installation and maintenance.

Education

PHD degree, National School of Engineering of Tunis — 01-2026
Thesis: Multi-Objective Optimization using Multi-Agent Federated Learning for the service of 6G networks.

Master's Degree, Sciences, Technologies, Santé, Mention Informatique, — 09-2023 – 12-2024
Université Paris Cité — Paris, France, Remote
- Joint degree with the Master's program at the National School of Engineering of Tunis
- Graduated with *High Distinction*

Master's Degree , Information Processing and Complexity of Living (TICV), — 09-2023 – 12-2024
National School of Engineering of Tunis — Tunis, Tunisia
- Ranked 1st out of 20 students.
- Thesis: Federated Learning for Early Diabetic Retinopathy Diagnosis

Telecommunications Engineering Degree, National School of Engineering of Tunis — 09-2021 – 06-2024, Tunis, Tunisia
- Ranked 1st out of 52 students.
- Specialized In Data Science for Embedded Communications (DASEC)
- Final Year Project: Forecasting Engine Automation (Infor Tunisia)

Two Year Undergraduate Program In Physics-Chemistry for the National Competitive Entrance Examination to Graduate Engineering School, University of Monastir — 09-2019 – 06-2021, Monastir, Tunisia

- Rank 113 out of 1300.

Languages

Arabic (Native Language), French (B2), English (B2), German (New)

Skills

Programming Languages (Python, C, C++, Java)

Data Science & Analytics (Different Python Libraries, SQL/PLSQL, Power BI, Tableau.)

Machine / Deep Learning & AI (TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Federated Learning (Flower, FedLab), AI Explainability (LIME, SHAP).)

Software, Backend, Mobile & Web Development (Flask, Spring Boot, JEE, MongoDB.)

Infrastructure (Linux, Docker, Kubernetes, Git)

Personal Projects

The Artificial Intelligence In The Service of Autonomous Cars — 09-2023 – 12-2023

- Used a Genetic Algorithm to design a bidirectional hyperdirective antenna network.
- Used YOLO V8 for Object detection and instance level segmentation using KITTI dataset.
- Realized an accuracy of 87%.

Chest X-ray (Pneumonia): Classification Model Using Transfer Learning — 06-2023 – 06-2023

- Used Transfer Learning in two different structures of CNN: ResNet and VGG
- Remarked that VGG is better than Resnet for Small Datasets.

Lip Reading of Tunisian Dialect using Machine Learning — 12-2022 – 03-2023

- Collected Tunisian DataBase (videos) and Preprocessed it.
- Extracted Features using OpenCV and DCT and Applied Some Features Selection Techniques.
- Applied machine Learning Models such as KNN and Random Forest.
- Attained an accuracy of 42%.

Certificates

Deep Learning Specialization , AWS Fundamentals Specialization , CDOSS Machine Learning With Python , Fundamentals of Deep Learning , Building Transformer-Based Natural Language Processing Applications , Programming Foundations with JavaScript, HTML and CSS , Foundations of Project Management , Interfacing with the Arduino

Volunteering

HUAWEI, Seeds for the future 2024 candidate — 07-2024 – 07-2024, Essaouira, Morocco

- Top 5 out of 28 teams from the North Africa region
- Learned and applied entrepreneurship and innovation skills.

IEEE ENIT SB, One year as an active member, and another year as a project manager — 10-2021 – 05-2023, Tunis, Tunisia

- Contributed to the success of various events as a member of the Logistics and Organization team.
- Appointed as the Ambassador and Organizer of the IEEE 16th event.
- Developed strong organizational, communication, and leadership skills

Awards

Achievement Award for Excellence., University of Tunis El Manar — 05-07-2024

Talan Tunisie Summer Camp , Top Performer, Talan — 31-07-2023

Passion

Competitive Programming, Reading Scientific Papers, Cooking Tunisian Foods, Football